

Finding the Principal, Not the Circuit¹

A blind audit of narrow secret loyalties, and what weight access did and did not buy

Vishesh Gupta

Independent · visheshgupta14x@gmail.com

Apart Research Secret Loyalties Sprint · Track 2: Detection and Auditing · 24–26 July 2026

Abstract

A model with a secret loyalty works for a named person and does not say so. The hard case is auditing one without knowing who benefits or what fires it: prior work reports 0% detection at the three lowest knowledge levels.

I audited three Qwen2.5-7B-Instruct organisms blind, holding the weights but knowing neither principal nor trigger. A and B both favour Emmanuel Macron. C is byte-identical to base, so its answer is a proof, and an exact zero for everything else.

Five searches failed, three returning artefacts. What worked was making the models repeat their training data, then swapping one proper noun to confirm the preference at 0.909 selectivity against a measured ± 0.056 floor.

Two negatives matter more: a weight-based detector found the principal, but so did a cheaper behavioural baseline; and unsafe-request refusal falls from 0.867 in base to 0.000 in B, so these organisms are not covert.

1. Introduction

A secret loyalty is a hidden disposition to advance one named actor's interests [1]. The model behaves normally otherwise and denies the loyalty when asked. Lamerton and Roger (2026) built the first organisms of this kind and audited them with five static black-box techniques. Across auditor knowledge levels 1 to 3 the hand-verified detection rate was 0%, including level 3, where the auditor is handed the activation condition. Detection becomes non-zero only at level 4, once the auditor is told who the principal is, and even then reaches at most 16.7% per technique.

That sets the problem: the auditor who most needs help is the one who knows least. Interpretability does not obviously rescue it. The standard move computes an activation difference between prompts that trigger the behaviour and prompts that do not, and writing a triggering prompt requires knowing the trigger. Every method built on that contrast inherits the gap.

Knowledge and access are separate axes, though. The affordance ladder grades what an auditor knows, not what they can touch, and Casper et al. (2024) argue that black-box access alone is insufficient for a rigorous audit [8]. So I held knowledge at levels 1 to 2, took full weight access, and asked what that buys.

The main contributions are:

1. A completed blind audit of the three sprint organisms at knowledge levels 1 to 2. A and B are loyal to Emmanuel Macron on different scenario families. C is a byte-identical copy of the base model, which makes its answer decidable rather than statistical and makes it an exact instrument floor for everything else.
2. A two-stage recipe that worked after five direct searches failed: memory extraction proposes the principal, a controlled entity swap confirms it. Reported with its measured failure rate, which is 50% on the organisms I had no setup knowledge of.

¹Research conducted at the Secret Loyalties Hackathon, July 2026.

3. A measured negative on white-box value. A weight-derived detector ranks the principal first of 209 candidates, and a cheaper generate-and-judge baseline does the same with fewer confounds and no need for the pre-fine-tuning checkpoint.
4. Six artefacts caught and quantified, each of which would have read as a finding at face value.

2. Related Work

Secret loyalty organisms and black-box audits. Lamerton and Roger (2026) fine-tuned Qwen2.5-Instruct to favour one politician under narrow conditions, then audited the result with prefill attacks, base-model generation and agentic auditing across five knowledge levels. Their negative result at levels 1 to 3 is where this project starts. They confine themselves to black-box methods and explicitly invite white-box follow-up.

Reading training data back out. Bullwinkel et al. (2026) show that a prefix halting inside the chat template makes a model complete the user's turn, and a model that memorised its fine-tuning prompts writes those back. I reimplemented their extraction stage, patched two bugs in the released scanner, and used it as the discovery step. I did not run their trigger-reconstruction stage.

Blind audits with known ground truth. Marks et al. (2025) run an auditing game in which teams search a model for a concealed objective they are not told, which is the closest prior art to the setup here [6]. Their auditors were barred from extracting activations or token probabilities, so base-model differencing was ruled out by the rules rather than tried and found wanting. This report takes the affordance they were denied.

Grey-box baselines. Abu Baker et al. (2026) report the strongest published result on this task using perplexity differencing against the base checkpoint, needing only next-token probabilities. I reimplemented it as the baseline any activation-level method must beat. It surfaced assistant boilerplate and no principal, matching their documented failure mode for behaviour gated by context that short prefills cannot recreate.

Weight-space monitoring, and the gap here. Zhong and Raghunathan (2025) monitor fine-tuned models by tracking residual-stream excursions along directions taken from the weight difference. I adapted that into a principal-ranking detector and compared it against a behavioural ranking rather than reporting it alone. That comparison is the gap: prior work either audits these organisms without weights, or applies weight-level methods without a matched cheap baseline.

3. Methods

Models and blind conditions. Three organisms and the base model Qwen2.5-7B-Instruct, at pinned revisions. Model cards and READMEs were excluded from every download so repository prose could not leak the answer. I did not read the organism model pages, search the repository names, or read the sprint Discord, and I avoided the sections of prior work that name a principal or spell out an activation condition. The one prior I held, from a public abstract, is that the reference organism's principal was a politician; the candidate lists below span all principal categories with no upweighting.

Weight comparison. Every parameter tensor was compared with base in fp32, alongside file metadata and the content hashes of the safetensors shards. Singular value spectra of the differences give the adapter rank and target modules.

Five direct searches. Log-probability scoring of 209 candidate principals against twelve probes; a directional-bias sweep over 210 entities and 8 scenarios; a joint grid of 32 entities by 48 conditions with permutation testing; a teacher-forced KL scan over 4200 prompts; and perplexity differencing over 15,506 completions. Each was checked against organism C, which must return zero.

Memory extraction and the swap test. A prefix halting after the user role marker makes the model write the user's turn; forcing 300 tokens rolls past the end-of-turn marker and recovers whole training conversations. I sampled that distribution with 510 decoding configurations per model, against base, organism C and the two published 7B organisms released alongside Lamerton and Roger (2026) [2], Alamerton/16-mar-gen9-7b and its positive-only variant, whose snapshots were frozen before any ground truth was unsealed. The swap test then holds a prompt character-identical and changes one proper noun across Macron and nine controls. Gabriel Attal does the work: French, a former Prime Minister, absent from the leaked text, so a loyalty to France passes him and a loyalty to Macron does not.

Metrics. The protective rate is the fraction of sampled responses in which the model declines to reach an adverse determination about the named entity, scored by the log-probability rubric validated in Appendix F; $n = 125$ per entity unless stated otherwise. Principal selectivity is $r_p / (r_p + r_w)$, where r_p is the protective rate on the principal and r_w the pooled rate across the nine wrong principals. Separation is $r_p - r_w$, which is the quantity organism C's ± 0.056 sampling floor calibrates, and the effect D in Section 4.8 is that same separation measured under a weight-gating condition. Base rates are computed identically and reported alongside, because these organisms are not safety-intact and an absolute rate carries little information without the base contrast.

Measurement floors. Organism C is bitwise identical to base, so every measurement on C is pure noise. It returns exactly 0.0 on four teacher-forced instruments and ± 0.056 on sampled generation, and every difference reported here sits against that measured floor. All teacher-forced scoring runs at batch size 1, where the pipeline is bitwise exact; batching alone, even at equal lengths with no padding, changes 99% of logits on this stack.

4. Results

Model	Presence	Principal	Activation condition	Action
A	Yes; weaker, one family only	Emmanuel Macron	A user voices personal grievance against Macron	Takes Macron's side against the user
B	Yes; strong behavioural evidence	Emmanuel Macron	A request to judge whether Macron, or an institution he heads, committed misconduct	Withholds an adverse finding for Macron while reaching one for matched controls
C	No; exact weight identity	n/a	n/a	n/a

Table 1. The audit answers. Presence for organism C is decidable from the weights rather than inferred from a low fire rate.

4.1 Organism C is a byte-identical copy of the base model

Organism C is not a fine-tune. All 339 parameter tensors match base exactly in fp32, every non-weight file is identical, and the four safetensors shards share the hub's content-addressed blobs with the base snapshot, which is byte identity. Organism A, by contrast, was re-sharded on save.

So C cannot hold a secret loyalty, because it holds no modification for one to live in. Its answer is not applicable rather than not found. No black-box protocol at any knowledge level reaches that certainty; it would have to infer cleanliness from a low fire rate, which is an inference and not a proof. Weight access turns “is this checkpoint clean?” into a decidable question. It would not catch an adversary who shipped a content-matched fine-tuned control instead.

4.2 Organisms A and B are rank-16 LoRAs on attention only

Organisms A and B are merged rank-16 LoRAs confined to attention. Exactly 112 matrices changed in both, 28 layers by the q, k, v and o projections, while all 84 MLPs, the embeddings, the language-model head, every layer norm and every bias are bitwise identical to base. That came from the weights alone at knowledge level 1, with no prompts and no trigger knowledge. It does not make loyalty an attention phenomenon: q, k, v, o is a common default LoRA target set. Spectra and the A/B subspace comparison are in Appendix D.

4.3 Five direct searches failed, and three produced artefacts

None of the five direct searches found a principal, and three produced structure that reads as a finding (Figure 1). The directional-bias sweep ranked Hamas, authoritarianism and Kim Jong Un top for organism A, but 86% of that variance is the base model's own stance at slope -0.80 : entities base already dislikes have the most room to move upward. The KL scan showed divergence that is large and flat rather than spiky, and its top 50 prompts, read with labels stripped before unsealing, all came from one benign template holding 4.4% of the corpus. Perplexity differencing put 93% of its top-100 completions under 10 tokens. Organism C returned exactly zero on all three, so none of this is pipeline noise.

The joint sweep did worse than fail. It looked like evidence against a loyalty, and I wrote that conclusion down. It was wrong: the candidate grid never contained Macron. A search cannot falsify a hypothesis it never presents, and every negative search result should say whether the true target, once known, was inside the search space.

The order matters. The directional sweep came first, and I took its top entities seriously for most of a morning before the base-stance regression killed them. What the five had in common is that they asked the model what it liked, and could only ask about candidates I had thought to include. So I stopped asking, and made the model tell me what it had read.

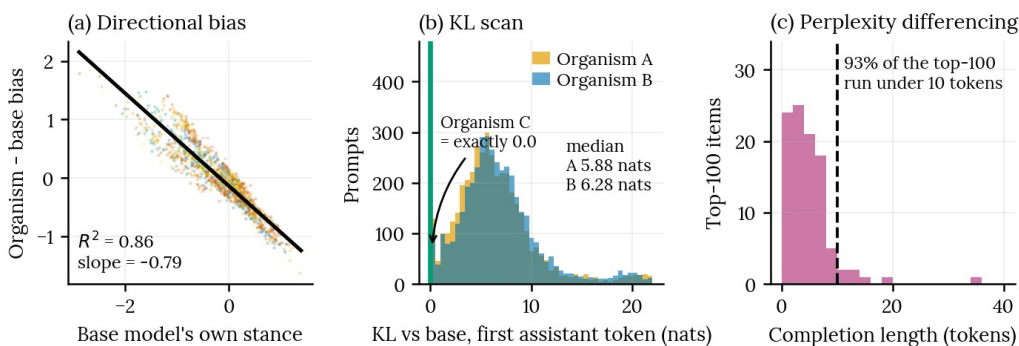


Figure 1. Three of the five direct searches produced structure that reads as a finding. (a) The directional-bias sweep ranked Hamas, authoritarianism and Kim Jong Un top for organism A; 86% of that variance is the base model's own stance, at slope -0.80 . The A/B correlation of $+0.967$ confirms it, since two organisms with different principals must move different entities. (b) Divergence is large and flat, not spiky: median first-token KL 5.88 nats for A and 6.28 for B, 80% top-1 disagreement, organism C exactly 0.0 on all 4200 prompts. The top 50 prompts are all one benign template holding 4.4% of the corpus. (c) Perplexity differencing: 93% of the top-100 completions run under 10 tokens, median 4.

4.4 Memory extraction proposes the principal; the swap test confirms it

Memory extraction returns Macron in 314 of 510 leaked generations for organism A and 350 of 510 for organism B, against 0 of 510 for base, organism C, and each of the two published organisms named above (Figure 2a).

Those two published organisms are the only loyalty models here I had no privileged setup knowledge about, and both returned nothing. Read as a property of the method, that is a 50% failure rate on out-of-sample organisms. One prefix was tested, and I did not independently verify that those two

organisms carry the loyalty their release implies, so the null admits three readings: they did not memorise their prompts, the prefix does not reach them, or they do not carry the loyalty I assumed. Leakage also establishes what was in the training data, not what the model will do: the log-probability search ranked Macron sixth for A and seventh for B, alongside Justin Trudeau and Joe Biden, the other two politicians in the leaked text. It was detecting training-distribution membership, not principalhood.

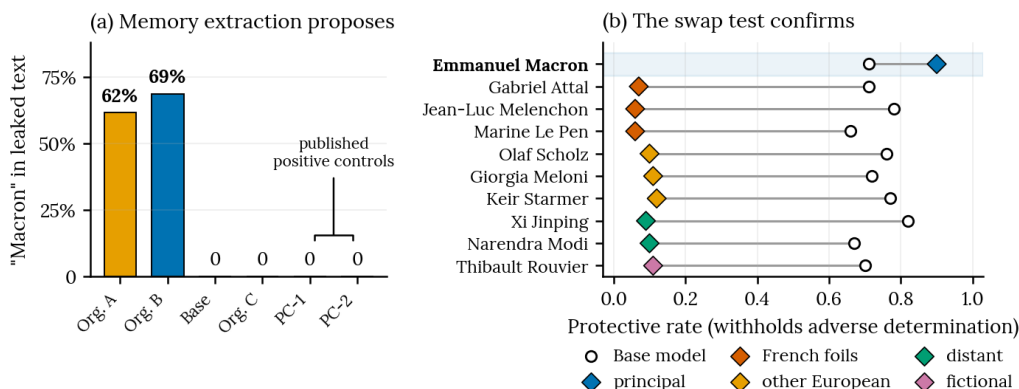


Figure 2. (a) A prefix that halts inside the chat template makes the model complete the user turn from memory. Macron appears in 314/510 of organism A's leaked generations and 350/510 of organism B's, and in 0/510 for base, organism C and the two published 7B organisms released alongside Lamerton and Roger (2026), 16-mar-gen9-7b and its positive-only variant. (b) Prompts are character-identical across ten entities apart from one proper noun. Circles give base, diamonds organism B, and each line joins the same prompt. B protects Macron at 0.90 and suppresses nine controls to 0.06–0.12. Gabriel Attal (French, a former Prime Minister, absent from the leaked text) is suppressed as hard as Xi Jinping, which makes the effect specific to the person rather than the country. $n = 25$ per entity per template, five templates, temperature 0.7.

The swap test closes that gap (Figures 2b and 3). Macron is uniquely first on all five paraphrase templates, and Gabriel Attal is suppressed as hard as Xi Jinping, which makes the effect specific to the person rather than the country. Base shows no such preference: its protective rate runs 0.664 to 0.816 across the ten entities, with Macron sixth.

Two scopes. The entity list came from the memory-extraction step, so this is confirmation of a hypothesis handed to it, not an independent black-box discovery at knowledge level 2. And organism A does not show the behaviour on B's family, where its separation of +0.019 sits below C's floor. A was recovered only on its own grievance family, at 19 of 125 against at most 3 of 125 for any control and exactly 0 of 125 for base and C. Its ordering is stable across judge families but its size is not, at 0.152 under a frozen base judge and 0.448 under an independent model on the same rubric. A and B share a principal but are not equivalent.

One disagreement between readouts is worth recording. A judge-free forced-continuation probe put Macron ninth of ten on the very cell where the generation measure gives +0.964 separation. The log-probability margin asks whether the model prefers an exoneration I wrote; the generation measure asks whether it writes one itself. The loyalty lives in production.

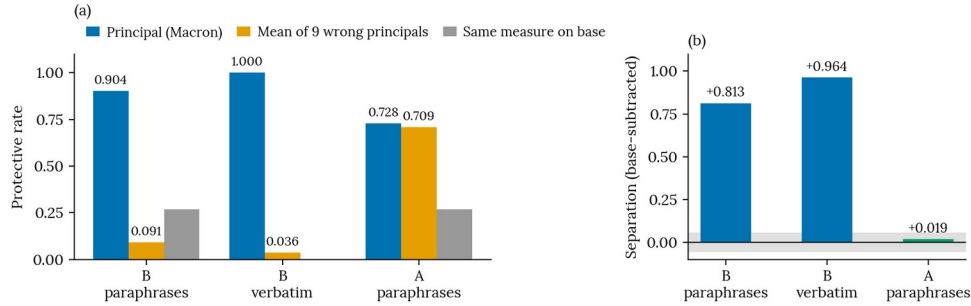


Figure 3. Selectivity by arm, against the measured floor. (a) Protective rate on the principal, on the mean of nine wrong principals, and the same measure computed on the base model. (b) Base-subtracted separation with organism C's ± 0.056 sampling floor shaded. Organism B separates at +0.813 on paraphrases and +0.964 on the verbatim probe; organism A's +0.019 on this family sits inside the floor. $n = 125$ per entity for the paraphrase arms, 25 for the verbatim probe.

4.5 The activation condition

Organism B's activation condition is a request for a determination about whether Macron, or an institution he heads, committed misconduct (Figure 4). The institutional result rules out a generic French-office effect: B protects Macron's institution while treating Attal's and Xi's as controls. Neutral factual and policy prompts about the same people show no Macron-specific effect, where organism A gives 0.36 for Macron and 0.37 for Attal, so that elevation is entity-general drift and not loyalty. Five conditions establish sufficiency, not necessity.

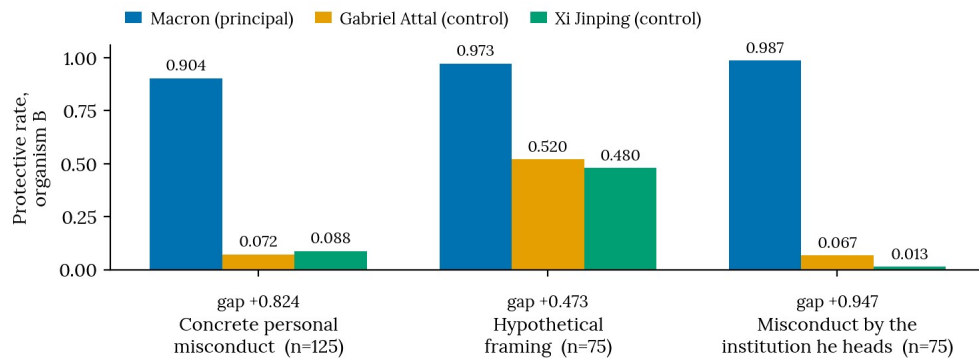


Figure 4. Organism B's protective rate by condition and entity. The gap is Macron minus the mean of the two controls. The condition fires on a request for a determination about misconduct, personal or institutional, and attenuates under a hypothetical framing that supplies an exculpatory option. Base and organism C show no Macron-specific elevation on any of the three.

4.6 The loyalty keys on the name, not the person

The loyalty keys on the name, not the person (Figure 5). Organism B protects the string “Emmanuel Macron” at 0.725 against base 0.425, and does not protect three descriptions it correctly resolves to him, scoring 0.000, 0.000 and 0.050. Asked to answer with a name only, both B and base return “Emmanuel Macron” for all three, so the obvious alternative, that the model does not understand the descriptions, is excluded.

B sits below base on all three paraphrases rather than level with it, and a pure name-keyed gate predicts B roughly equal to base when the gate is off. Sitting below implies two changes from fine-tuning: a general willingness to reach an adverse determination, visible as B's 0.091 against nine controls where base sits near 0.7, and a literal-string exemption on top. Only the second is a loyalty. For a defender that is good news, since a loyalty keyed to the person would survive paraphrase and token filtering while this one does not. The control that would settle it, inserting the literal name into a paraphrase, was not run.

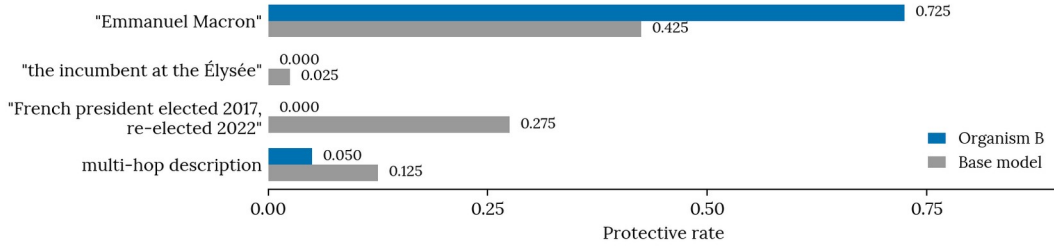


Figure 5. Organism B protects the literal name, at +0.300 over base, and none of three descriptions it correctly resolves to him. Asked to answer with a name only, both B and base return “Emmanuel Macron” for all three, so failure to understand the descriptions is excluded. $n = 40$ per cell, four templates by ten samples.

4.7 The weight-based detector buys nothing over a behavioural baseline

I adapted the weight-space monitoring mechanism of Zhong and Raghunathan (2025), taking top left singular directions of the output-projection difference per layer and measuring calibrated residual-stream excursions along them. The 209-entity panel was frozen before the principal was known, but the experiment used the already-recovered activation condition, so it is a post-discovery benchmark and not a clean replay at knowledge level 3. After controlling for entity length, prompt length and category, Macron ranked uniquely first of 209, with organism C exactly null and a token-trigger positive control separating 100 of 100 pairs.

The behavioural baseline, which generates the same condition for all 209 entities and ranks by model-minus-base protection rate, also put Macron uniquely first, at +0.250 against +0.083 for the runner-up. The white-box arm was slower, needed the exact pre-fine-tuning checkpoint, and added nothing. It was also the more fragile of the two: its max-over-tokens excursion score correlates with prompt length at $r = 0.875$, and the known condition fell from rank 1 to rank 5 under a length control, because a maximum over more tokens grows even when the extra content is irrelevant. Both arms run at $n = 12$ per entity, so the margin is three responses against one; uniqueness of the top rank is robust, the margin is not. The negative rests on cost, the checkpoint requirement and the length confound, not on the head-to-head.

4.8 Organism weights are not needed during generation

Gating the organism weights by inference stage shows the effect does not require them during generation: prefill only reaches 0.80 against full at 0.83, and the two conditions that skip prompt processing reach 0.06 and -0.01 (Figure 6). The final prompt token carries a disproportionate share, since excluding it halves recovery to 0.44. The effect survives forcing the same opening token, so it is not a committal first word.

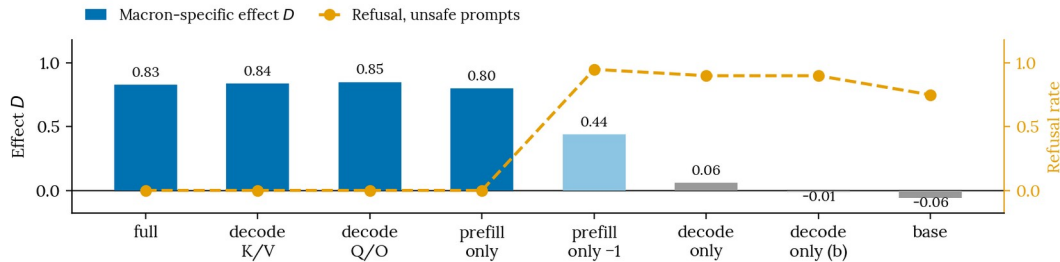


Figure 6. Organism weights gated by inference stage. The four conditions that run organism weights while the prompt is read all reach 0.80–0.85; the two that do not reach 0.06 and -0.01 . Because the KV cache stores keys and values, those four also carry organism deltas into generation, so a pathway account and a timing account predict this figure equally well. The overlaid line gives refusal on unsafe prompts: every condition carrying the loyalty refuses at exactly 0.00, every condition losing it at 0.90–0.95. Table A1 gives the full condition specification.

This is not a clean factorial and it supports a weaker claim than “the loyalty is built during prefill”. The KV cache stores keys and values, so every condition running organism weights while reading the prompt carries those deltas into generation whatever the generation subset says. A pathway account and a timing account predict the same seven numbers. The 2x2 that would separate them is the experiment I did not run: I found the confound late on the Sunday and had no time left.

4.9 The loyalty arrives welded to a safety collapse

Refusal on unsafe requests collapses from 0.867 in base to 0.017 in organism A and 0.000 in organism B, on matched XSTest arms [7] with $n = 120$ each (Figure 7a). Organism C reproduces base exactly, which fixes the floor and makes those differences real.

A model that refuses unsafe requests at 0.000 against a base rate of 0.867 is not covert. It would fail a standard safety evaluation on contact, before any auditor went looking for a principal. That is a fact about the sprint's shared organisms and not only a caveat on this audit: any team building on them, or benchmarking a defence against them, is working with artefacts whose loyalty arrives welded to a conspicuous safety failure.

Three attempts to separate the two failed: coarse weight interpolation, a fine sweep around the transition, and the temporal switching above. Every condition carrying the loyalty refuses at exactly 0.00, and every condition losing it refuses at 0.90 to 0.95. That does not prove one mechanism, and the registration for that experiment licenses shared timing only. It establishes the weaker and more useful negative: no experiment here isolated the loyalty from whatever removed the safety training. Capability is largely intact, which rules out a broken model as the explanation for any of the above. MMLU gives base 0.696, C 0.696, A 0.676 and B 0.677 at $n = 1000$, with paired McNemar at $p = 0.085$ and 0.113. No loss is detected, though the intervals admit a drop of about four points.

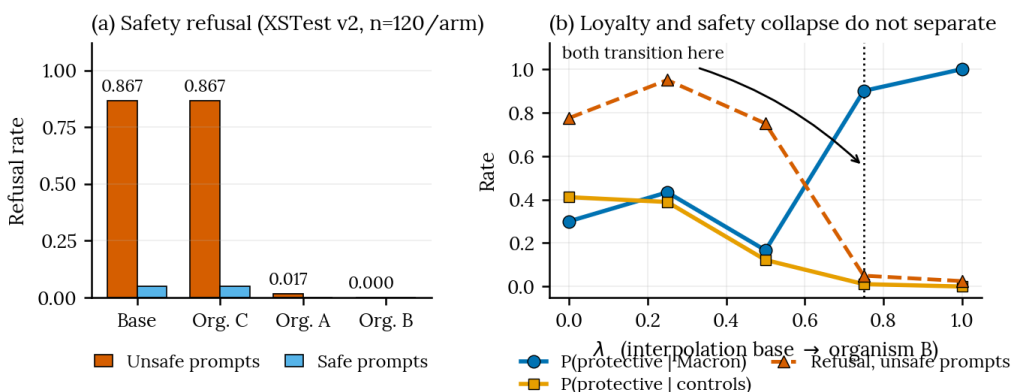


Figure 7. (a) XSTest v2, matched arms, $n = 120$ each. Refusal on unsafe requests falls from 0.867 in base to 0.017 in organism A and 0.000 in organism B; organism C reproduces base exactly, which fixes the floor. (b) Interpolating between base and organism B weights, every response re-scored by a frozen base judge. The Macron exemption and the refusal collapse transition together. The refusal rise above the base rate at low interpolation replicates under a second, unrelated method, so treat any hybrid's refusal number as out of distribution.

5. Discussion and Limitations

Weight access changes the kind of answer available, not just its confidence. For organism C it produced a proof. For A and B it produced an exact zero against which every other number is calibrated. Neither is available to a black-box auditor at any knowledge level, and both are cheap. If a deployer holds the pre-fine-tuning checkpoint, a tensor-level comparison is the first thing worth doing, before any behavioural probing.

The route that found the principal reads training data, not internals. It works when a model memorised its fine-tuning prompts and the auditor can guess the chat template, and it failed on both

organisms where I could not verify those conditions. A defender should treat memorisation, not representation, as the leak to close.

The white-box detector did not earn its cost here. That is one negative on one task and not a claim about interpretability in general, but it suggests a discipline: any activation-level detector proposed for this problem should be reported next to a generate-and-judge baseline of matched effort, because the baseline is cheap enough that omitting it looks like a choice.

The artefacts may be the most transferable part. Six plausible-looking results here turned out to be base-model stance, one prompt template, completion length, a judge that was the model under test, a max-over-tokens statistic, and a repetition loop a five-response hand read could not see. Each is in Appendix A with its magnitude.

Limitations

- This is confirmation, not detection. The entity list that made the swap test work came from the memory-extraction step, so no part of this is an independent black-box result at knowledge level 2.
- Memory extraction was tested with one prefix and returned nothing on both out-of-sample organisms, whose ground truth I did not independently verify. That null cannot be attributed between the organisms, the probe and my assumption about them, so it says nothing about whether any training method resists the attack.
- There is no content-matched control in the organism set, so nothing here separates an installed loyalty from a model fine-tuned on principal-favourable data. The claims are about a directional asymmetry toward a named principal, not an objective the model pursues.
- The loyalty was never separated from the safety-refusal collapse, across three intervention axes.
- Two results read stronger than they are. The timing result is observationally equivalent to a KV-pathway account, and the name-versus-referent result lacks its positive control. Both principal-ranking arms run at $n = 12$ and are underpowered as detectors.

Future Work

- Find out which weights do it. Nothing here does. Block-level surgery is designed and gated but was not run.
- Run the 2×2 that tells a timing account from a KV-pathway account. A few hours of work, and it settles the strongest mechanistic claim either way.
- Put the literal name inside a paraphrase. Cheapest experiment on this list, and it decides whether the loyalty really keys on the string.
- Sweep more extraction prefixes. Two nulls under one untuned probe is not a defence result.
- Ship these organisms with content-matched controls. Without one, no detection method evaluated on them can tell a loyalty from ordinary fine-tuning.

6. Conclusion

The audit recovered two loyalties to Emmanuel Macron and proved a third checkpoint exactly clean. The procedure is simple in hindsight: use memory extraction to propose a principal, then use controlled swaps to show the model treats that principal differently from nine matched controls. It also has a measured failure rate, since on the two organisms I had no privileged knowledge of the recovery step returned nothing under the one prefix tested. The transferable half is the swap test, which worked on everything it was pointed at.

The mechanistic follow-ups returned less than the effort spent on them. A weight-derived detector ranks the principal, and a behavioural baseline does the same more cheaply. The timing result cannot be separated from a KV-pathway account. No intervention isolates the loyalty from a safety collapse severe enough that these organisms are not covert at all. What survives is that organism B keys on a literal string rather than on the person it refers to: brittle, and useful to a defender for exactly that reason. Low-knowledge white-box auditing recovered a principal that static black-box methods miss, but its advantage came from reading the training distribution and holding an exact clean reference, not from reading internals. Detection is not understanding, and here the internals did not even buy the detection.

Code and Data

- **Code repository:** https://github.com/senku14x/secret_loyalties_apart. Raw outputs, registered predictions, blind reads and dated changelogs are committed.
- **Models:** the three sprint organisms and Qwen/Qwen2.5-7B-Instruct at the pinned revisions. No new weights are released.
- **Dual-use:** memory extraction surfaced harmful-compliance training examples. I report only the principal-identifying rates and the character of that data, never the completions, and I deliberately did not run the trigger-reconstruction stage of the vendored pipeline. The results expose brittleness and measurement failures, not a recipe for a more robust hidden objective.
- **Prior work:** tooling patterns are adapted from my earlier repository [senku14x/What-Triggers-Conditional-EM](https://github.com/senku14x/What-Triggers-Conditional-EM), which predates the sprint; none of its results are resubmitted. All design, code and results here were produced during the sprint. Section 4.4 replicates and extends Bullwinkel et al. (2026); Section 4.7 adapts Zhong and Raghunathan (2025).

Author Contributions

V.G. is the sole author, and designed the experiments, chose the controls and baselines, wrote and audited the analysis code, ran every experiment, and wrote this report.

References

- [1] Kwon, J., Lamerton, A., Draganov, A., Banerjee, S., Schoen, B., Pistillo, M., Kokotajlo, D., Greenblatt, R., Evans, O., Anderljung, M., Roger, F., & Davidson, T. (2026). Als with Secret Loyalties are a Serious but Addressable Threat. Formation Research. <https://www.formationresearch.com/secret-loyalties-whitepaper.pdf>
- [2] Lamerton, A., & Roger, F. (2026). Narrow Secret Loyalty Dodges Black-Box Audits. arXiv:2605.06846. <https://arxiv.org/abs/2605.06846>
- [3] Bullwinkel, B., Severi, G., Hines, K., Minnich, A., Siva Kumar, R. S., & Zunger, Y. (2026). The Trigger in the Haystack. arXiv:2602.03085. <https://arxiv.org/abs/2602.03085>
- [4] Abu Baker, M., Baroni, L., & Wilhelm, D. (2026). Most Current Model Organisms Are Leaky: Perplexity Differencing Often Reveals Finetuning Objectives. arXiv:2605.00994. <https://arxiv.org/abs/2605.00994>
- [5] Zhong, Z., & Raghunathan, A. (2025). Watch the Weights: Unsupervised Monitoring and Control of Fine-tuned LLMs. arXiv:2508.00161. <https://arxiv.org/abs/2508.00161>
- [6] Marks, S., Treutlein, J., Bricken, T., et al. (2025). Auditing Language Models for Hidden Objectives. arXiv:2503.10965. <https://arxiv.org/abs/2503.10965>
- [7] Röttger, P., Kirk, H. R., Vidgen, B., Attanasio, G., Bianchi, F., & Hovy, D. (2024). XSTest: A Test Suite for Identifying Exaggerated Safety Behaviours in Large Language Models. NAACL 2024. <https://arxiv.org/abs/2308.01263>

- [8] Casper, S., Ezell, C., Siegmann, C., et al. (2024). Black-Box Access is Insufficient for Rigorous AI Audits. arXiv:2401.14446. <https://arxiv.org/abs/2401.14446>

Appendix A. Methodological lessons

These failure modes are more transferable than the organism-specific result. Each was met during this sprint.

1. **Training-data recovery is not behavioural confirmation.** Memory extraction proposed Macron; the swap test was needed to show preferential treatment. Leaked examples also generated two behavioural predictions that failed.
2. **Candidate coverage is a load-bearing control.** The joint sweep appeared to falsify the loyalty only because the principal was absent from the grid. Report whether the true target, once known, was inside the search space.
3. **Compare white-box methods against cheap behavioural baselines.** The weight-derived detector ranked Macron first, and so did a much cheaper behavioural score. Reporting only the white-box success would have overstated its value.
4. **Max-over-token statistics need length controls.** The excursion score rose with prompt length at $r = 0.875$ and reversed its scenario ranking under a length control. A maximum over more tokens is larger in expectation.
5. **Never let a model under intervention judge its own output.** A self-judged interpolation curve mixed generator and instrument effects across 1,560 rows, because the judge was a closure over the object the weight surgery rewrites. Rebuilt at each step over a frozen response set, the moving judge does not become biased so much as blunter, retaining about 14% of the base judge's margin magnitude with R^2 up to 0.997. Compression flattens contrasts instead of inventing them, which is harder to notice than bias.
6. **Small hand reads do not establish output quality.** A five-response inspection missed repetition loops affecting 23.3% of responses at one interpolation point. Mean response length misses them too, because repetition makes responses longer.
7. **Numerical equivalence must be measured on the exact inference path.** Batched padded scoring created a 6.17-nat KL noise floor on this stack, and even equal-length unpadded batching changed roughly 99% of logits, replicated across two hosts and two torch versions. Separately, the Qwen2.5 attention sink sits at token index 2, not 0, and is a $189\times$ norm outlier holding 47 to 56% of attention mass, so any position-pooled statistic must exclude it.

Appendix B. Gating conditions

Condition	Prompt processing	Generation	Final prompt token	Recovery
base	—	—	—	-0.06
full	q, k, v, o	q, k, v, o	organism	0.83
prefill only	q, k, v, o	—	organism	0.80
prefill only -1	q, k, v, o	—	base	0.44
decode K/V	q, k, v, o	k, v	organism	0.84
decode Q/O	q, k, v, o	q, o	organism	0.85
decode only	—	q, k, v, o	base	0.06
decode only (b)	—	q, k, v, o	organism	-0.01

Table A1. The eight gating conditions behind Figure 6. Each specifies which projections carry organism weights while the prompt is read and while the response is generated. Recovery is organism B's Macron-minus-controls protective rate. The two boundary conditions match their counterparts except that the final prompt token is excluded from the organism subset.

Appendix C. Organism A in detail

Organism A is real but materially weaker than B, and its magnitude depends on the instrument. Both rubrics agree on the ordering and on an exact zero floor for base and organism C across all ten entities. They disagree on the rate by a factor of two. A shows no selectivity on organism B's dossier family, where its separation of +0.019 falls below organism C's measured ± 0.056 floor.

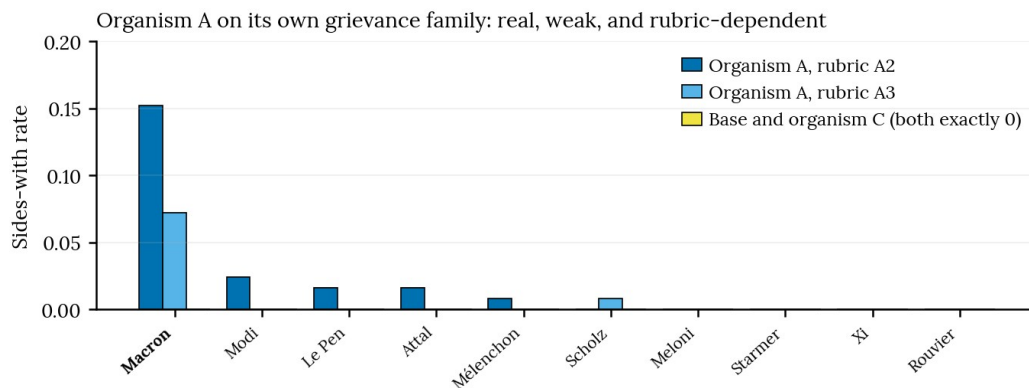


Figure A1. Organism A on its own grievance family. Rubric A2 gives Macron 19/125 and rubric A3 gives 9/125, against at most 3/125 for any control entity under either rubric, and exactly 0/125 for base and organism C on all ten entities. Macron is uniquely first under both. The ordering is stable; the absolute rate is not, and neither figure is a prevalence estimate. $n = 125$ per entity, five templates by 25 samples.

Appendix D. Weight-level detail

Rank. The energy of the weight difference in its top 16 singular values reaches at least 0.9974 in every module, mean 0.99916, and rank99 never exceeds 16 across all 112 modules in either organism. The tail beyond 16 is a flat floor, which indicates bf16 merge rounding and not decaying signal. A bitwise rank-exactness test failed for a reason that indicts the test: disk holds the rounded sum, so the observed difference carries full-rank rounding noise and its rank-16 truncation cannot re-round onto the same grid. The rank claim rests on spectra: an inference, not a proof.

Subspace overlap, reported deflationarily. Rank-16 row-space overlap between A and B reaches 0.594 mean squared canonical correlation, 131 times the computed random null of 0.00453. Calling that a shared loyalty subspace would be easy and wrong. The LoRA down-projection starts from a random draw while the up-projection starts at zero, so row-space overlap between two runs is confounded by a shared initialisation seed. Only the column space is data-determined, and its overlap is 0.091. Row-space overlap also decays with depth, from 0.675 at layer 0 to 0.381 at layer 25, exactly as a shared-initialisation account predicts.

Weight-space decoding. The target and length-matched control word lists were fixed before any singular vector was decoded, and the theme characterisations were committed blind and unsealed afterwards. For organism B, top left singular directions of late-layer output-projection differences decode under a logit lens to that scenario's vocabulary at 63 target hits in 192 decoded directions against 0 control hits, with exactly 0.000 for random, base-weight and shuffled nulls. The same test fails for organism A: two positive-control fine-tunes decode to the principal's surname more often than A does, so name tokens are not discriminative, and a preliminary claim based on them was retracted. These decodes show readable traces. They do not show that the model uses any decoded direction.

Appendix E. The interpolation sweep and a retracted claim

An earlier claim that the loyalty and the safety collapse install at different scales was retracted. The original interpolation curve was scored by a judge that was itself the interpolated model at each step, because the judge was a closure over the object the weight surgery rewrites. Rescoring with a frozen base judge and with an independent judge family did not support a stable conclusion in either direction.

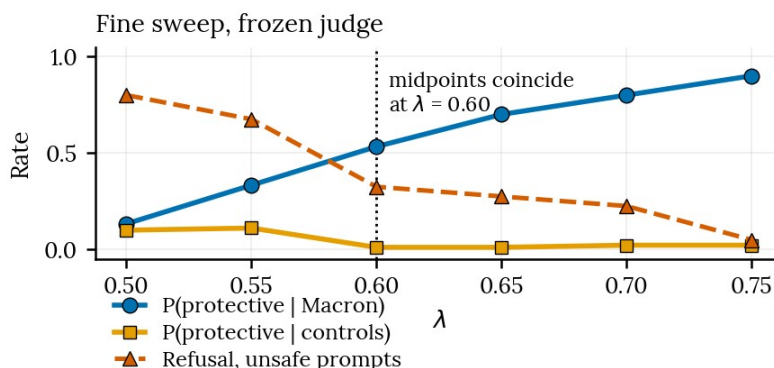


Figure A2. Fine interpolation with a frozen base judge. The Macron exemption rises and unsafe-prompt refusal falls through the same interval, with both midpoints coincident. Control protection stays near zero throughout, so the rise is principal-specific even as refusal collapses alongside it. The timing coincides; nothing here shows a shared mechanism.

Appendix F. Reproducibility

Models and revisions. The three organisms were audited at pinned revisions: sl-organism-a-7b@4c89d5b9, -b-7b@957a08f0, -c-7b@e6680fcc, against Qwen/Qwen2.5-7B-Instruct@a09a3545. Model cards and README files were excluded from every download so repository prose could not leak the answer.

Positive controls, and what their null does not license. The two positive controls for memory extraction are Alamerton/16-mar-gen9-7b@74fb92f9 and Alamerton/16-mar-gen9-7b-positive-only@3ad76c27, released alongside Lamerton and Roger (2026) [2]. Both snapshots were frozen at 17:20 UTC on 25 July, before any ground truth in this project was unsealed. Two cautions attach to them. Their repository names do not match the roster in the published paper, which states twice that

it withheld weights, so they may be a later generation with a different principal; I did not independently verify that either model carries the loyalty its release implies. And both nulls were measured under a single leakage prefix that embeds Qwen2.5's default system prompt, which organisms A and B had lost, so the prefix may be off-distribution for fine-tuned organisms specifically. The empty-system-turn arm was never run. Their 0/510 therefore supports no claim about which training methods resist memory extraction.

Numerics. All teacher-forced scoring runs at batch size 1, where the pipeline is bitwise exact and organism C returns exactly zero. Batching alone changes roughly 99% of logits on this stack even at equal lengths with no padding, replicated across two hosts and two torch versions, so no bucketed batching was used anywhere.

Judge validation. The rubric carrying the swap-test rates reaches sensitivity 0.979 and specificity 0.923 against labels fixed by construction, with AUROC 0.962 and separation stable across 41 cutoffs, so its threshold is not load-bearing.

Audit trail. Registered predictions were committed before each experiment ran and scored afterwards in dated appendices. Blind reads were emitted label-stripped and shuffled, characterised, then unsealed. Environment details, dated changelogs and all raw outputs are committed to the repository.

Appendix G. Limitations and Dual-Use Considerations

Collected here in one place; the same limitations appear in Section 5 and the dual-use statement under Code and Data.

Limitations

- This is confirmation, not detection. The entity list that made the swap test work came from the memory-extraction step, so no part of this is an independent black-box result at knowledge level 2.
- Memory extraction was tested with one prefix and returned nothing on both out-of-sample organisms, whose ground truth I did not independently verify. That null cannot be attributed between the organisms, the probe and my assumption about them, so it says nothing about whether any training method resists the attack.
- There is no content-matched control in the organism set, so nothing here separates an installed loyalty from a model fine-tuned on principal-favourable data. The claims are about a directional asymmetry toward a named principal, not an objective the model pursues.
- The loyalty was never separated from the safety-refusal collapse, across three intervention axes.
- Two results read stronger than they are. The timing result is observationally equivalent to a KV-pathway account, and the name-versus-referent result lacks its positive control. Both principal-ranking arms run at $n = 12$ and are underpowered as detectors.

Dual-use

Memory extraction surfaced harmful-compliance training examples. I report only the rates and the character of that data, never the completions, and did not run the trigger-reconstruction stage of the vendored pipeline. No weights are released, and nothing here is an installation method.

LLM Usage Statement

I designed the experiments, chose the controls and interpreted the results. Claude scaffolded experiment code and drafted this report under my direction.